

Rev 12 Errata - The Welch Labs Illustrated Guide to AI



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Location	Description
p. 18–19, Fig. 1.10	Column headers "Initial Dials" and "Initial Output" should read "Dials" and "Output."
p. 19, Fig. 1.9, Caption	"paaterns" should be "patterns."
p. 19, Fig. 1.10, Task 2, Step 3	The target input is negative and the output is positive, so all switched-on knobs should be turned down. The bias knob in the update column shows a red (up) arrow; it should be a green (down) arrow. The knob positions in the following step are correct.
p. 19, Fig. 1.10, Task 2, Step 7	The target output is negative and the actual output is positive, so all switched-on knobs should be turned down. The update column shows one red (up) arrow; all arrows should be green (down). The knob positions in the following step are correct.
p. 20, Fig. 1.13	"All dials set to zero make for zero" should be "All dials set to zero make for zero."
p. 21, Lines 2–3	Line 2 uses the inequality $0 < w_1x + w_2y + b$, but line 3 uses " $=$ ". Line 3 should also use " $<$ " for consistency.
p. 21, Para. 1	If $w_2 < 0$, the inequality reverses when solving for y . The boundary line equation is correct, but which side is "positive" depends on the sign of the coefficients.
p. 23, Task 2	The top-right dot representing the point (1, 1) should be colored yellow.
p. 24, Fig. 1.16	The step label and prediction read "Dog"; both should read "Cat."
p. 24, Fig. 1.16, Steps 2, 3, 5	The weights images appear incorrect and should be replaced.
p. 28, Para. 1	A period is missing at the end of the sentence "...we need to choose a specific target value."
p. 29	The $\hat{y}$ equation near the bottom of the figure is written twice

Location	Description
p. 30, Fig. 1.30	The equations under both "No Activation Functions" and "With Activation Functions" omit the x_2 term.
p. 34, Fig. 1.29, Caption	"Perceptron Learning Rule Here" is missing a closing period.
p. 34, Footnote 9	"when it's output is greater" should be "when its output is greater."
p. 36, Fig. 1.32	The final neuron in the network diagram is labeled $b_2^{(1)}$; it should be $b_1^{(2)}$.
p. 37, Para. 2	The period in "did for a single neuron. (Figures 1.22, 1.23) Following" should come after the closing parenthesis, not before it.
p. 38, Para. 5	"it's enormous network of neurons" should be "its enormous network of neurons."
p. 39, Fig. 1.35, Caption	"this exactly what GPT-3 is!" should be "this is exactly what GPT-3 is!"
p. 64–65, Problem 1.15	The formula for the second neuron is missing the x_2 term in the table header, in $f_2(x_1, x_2)$ above the table, and in $f(x_1, x_2)$ on p. 64.
p. 65, Problem 1.15	The biases in the perceptron diagram and parameter list have incorrect sub/superscripts in three places.
p. 73, Para. 1	"in this case 1.76" should be "in this case 1.75," consistent with Fig. 2.11.
p. 77, Fig. 2.18	The closed empty loop represents not only a minimum but also potentially a maximum.
p. 108, Para. 3	"returnsc" should be "returns a."
p. 110, Para. 3	$m_3 = 1$ should be $m_3 = -1$.
p. 116, Para. 1	When computing the partial derivative for Paris, $dL/dh_2 = 0.09 - 1$ is written as 0.91; it should be -0.91 .
p. 147, Para. 3	"panel E of Figure 4.2" should be "panel E of Figure 4.1."
p. 148, Fig. 4.4	Panel labels A through D are missing and should be added.
p. 151, Fig. 4.6, Caption	Several typos are present in the caption.
p. 152, Fig. 4.10	The x- and y-axis labels are swapped.
p. 153, Footnote 3	"need to differentiable" should be "need to be differentiable."
p. 154, Fig. 4.11	The node labeled $h_1^{(1)}$ on the network diagram should be $h_2^{(1)}$.
p. 160, Fig. 4.17	The caption refers to "Panels B and E," but no panel E exists, only panel B, in the figure. The panels only range from A through D.
p. 167, Para. 2	"this these regions tiny around around" should read "creates all these extra little polygons around our town's borders."
p. 201, Fig. 5.14	The second layer label reads "mixed4b" twice; the second instance should be "mixed4c."
p. 203, Para. 4	"we can to visually walk through embedding space" should be "we can visually walk through the embedding space."

Location	Description
p. 203, Para. 4	"such as direction in embedding space that amazingly correlate" should be "such as a direction in embedding space that amazingly correlates."
p. 205–207, Fig. 5.18	The figure on p. 207 is numbered 5.18, duplicating the figure on p. 205. It should be renumbered Fig. 5.20.
p. 225, Para. 1	A period is missing at the end of the sentence "...the OpenAI team released GPT-4."
p. 279, Fig. 8.10	The last column of QK^T should be highlighted in red, consistent with the multiplication by the last column of K^T .
p. 366, Exercise 3.7	$m_2 = 1 - 0.1(2.33) = 0.676$ should be $m_2 = 1 - 0.1(2.33) = 0.767$.